

Task Initiation: Why 'You Wanted to Do It but Didn't Start' Does Not Identify the Cause

Human-readable companion to Task-Initiation Failure as an Underidentified Endpoint: A Framework for Distinguishing Sources of the Goal-to-Action Gap

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

A person can know what needs to be done, endorse the goal, understand the consequences of delay, possess the needed ability, and still fail to begin. The visible endpoint - an intended task that did not start - does not tell us which part of the goal-to-action transition failed. The Squirrel manuscript treats strict non-initiation as an underidentified endpoint rather than as proof of weak motivation, poor executive control, low reward sensitivity, or any other single cause. It proposes four candidate functional contrasts: whether an executable next action was constructed (F1), whether the target policy locally won against alternatives (F2), whether onset differs when it must be self-generated versus externally supported after the target policy is selected (F3), and whether entering a new task configuration carries transition-sensitive cost (F4). These are not established modules, traits, or biological mechanisms. The framework orders perturbations and measurements so that candidate explanations can be separated, nested, merged, dropped, or replaced by lower-dimensional accounts. Post-initiation execution failure is kept outside the initiation endpoint. The proposal builds on established intention-behavior, implementation-intention, action-control, control-allocation, task-switching, and target-engagement work. Its residual contribution is an ambiguity-reduction architecture for asking what a failure to start actually licenses us to infer.

1. The human problem: knowing and intending are not the same thing as beginning

The puzzle behind Squirrel is familiar: a person can say exactly what needs to be done, agree that it matters, predict the consequences of delay, and still not begin. In other cases the opposite problem appears - once an activity has started, stopping can be unusually difficult. The final manuscript narrows its scientific target to the first problem: strict task initiation.

Everyday explanations often jump directly from the endpoint to a trait or motive. 'You did not start, so you must not have wanted it enough.' 'You procrastinated, so the problem is motivation.' 'You knew and still did nothing, so the problem is defiance or self-control.' Those interpretations may be correct in a particular case, but the observed endpoint does not identify them.

The manuscript formalizes the minimal observation as $G = 1$ and $Y \wedge I = 0$. $G = 1$ means the superordinate goal is represented and endorsed at the start of the bounded episode. $Y \wedge I = 0$ means the person did not cross a predeclared objective initiation criterion, such as the first valid task-directed action. From those two facts alone, weak motivation does not follow.

2. Why the endpoint is underidentified

Several different internal organizations can produce the same outward result. The person may not yet have converted an abstract outcome into a concrete next action. The target action may be available but fail to win against competing policies. The target policy may already be selected while self-generated onset remains delayed. Or entry into the target task configuration may impose a transition-sensitive cost. A failure that occurs after onset is a different endpoint again.

This is observational equivalence. Two candidate models can both predict non-initiation under the baseline condition even though they place the bottleneck in different functional locations. The endpoint therefore defines an equivalence class rather than a unique mechanism.

The framework becomes informative only when an intervention causes candidate models to diverge in what they predict. Even then, predictive divergence is not enough for strong causal attribution. A cue, prompt, or supplied next step may alter arousal, expectancy, perceived efficacy, preparation, attention, or policy preference at the same time. Stronger interpretation requires valid manipulation, proximal measurement where possible, and better out-of-sample discrimination than simpler alternatives.

3. The episode must be bounded before the factors are tested

The manuscript analyzes a bounded transition from an endorsed superordinate goal to observable task-directed onset. The focal person and the controlled task environment are both part of the declared analysis. The framework does not assume that every determinant lives inside the person.

The most important episode variables are separated because they answer different questions. G marks superordinate goal representation or endorsement. A is a proximal measure of whether a local executable action has been specified. P is local target-policy selection or choice propensity. $Y^{\wedge}I$ says whether initiation occurred. $T^{\wedge}I$ records initiation latency when onset occurs. $Z^{\wedge}E$ is a downstream execution outcome after onset.

The baseline class $G = 1$, $Y^{\wedge}I = 0$ deliberately does not assume $P = 1$. That matters because a person can endorse an overarching goal while a different local policy wins in the moment. Later onset contrasts are tested only after $P = 1$ is sufficiently established and an executable action is available. The ordering is analytical and evidentiary, not a claim that the brain implements four serial biological boxes.

4. F1: did the goal become an executable next action?

An abstract goal can be clear without specifying what to do next. 'Finish the project' identifies an outcome but contains less local action information than 'open the file and revise the first paragraph' or a cue-linked implementation intention. Established implementation-intention research already shows that linking situations to goal-directed responses can improve enactment.

F1 names the candidate distinction between representing an outcome and constructing an action executable in the current state. It can be written compactly as $G_{\text{abstract}} \rightarrow a_{\text{local}}$. The scientific point is not that a new action-construction mechanism has been discovered. It is that failure to construct a local action is logically different from assigning low value to the outcome.

Supplying the next step is therefore only an F1-sensitive manipulation, not an F1 diagnosis. A supplied action can also increase perceived efficacy or make the target policy more attractive. The framework requires a separate measure of action specification A and simultaneous measurement of policy-selection variables before an effect can be attributed to an independent F1 role.

5. F2: does the target policy win against what the person could do instead?

Once candidate actions exist, they still compete. The long-term importance of a goal does not guarantee that its local policy wins at this moment. Continuing the current activity, avoiding a difficult transition, waiting, choosing another task, or taking an alternative route can all compete with the target.

The manuscript expresses the comparison as a difference in local action values: $\Delta Q = Q(\text{target policy}) - Q(\text{competing policy})$. The exact mechanism is left to established control-allocation and decision theories. F2's job is narrower: distinguish superordinate goal endorsement from the local selection result.

An F2-sensitive test therefore needs evidence that the manipulation changed local target-policy preference or choice propensity P before it interprets the initiation effect. Because F1 manipulations can change efficacy and preference, F1 and F2 are explicitly allowed to nest or merge if proximal measurements show that the apparent action-construction effect is carried by policy selection.

6. F3: selected policy, but self-generated onset differs from supported onset

The most tempting part of the framework is also the one the manuscript treats most cautiously. Even after the target policy is selected, self-generated onset may differ from externally supported onset. External structure - a cue, instruction, prompt, or other onset support - can sometimes reduce initiation delay.

F3 is intentionally defined as an operational contrast rather than a discovered latent mechanism. The comparison is made only after $P = 1$ has been observed or stabilized and an executable action exists. The relevant question is whether onset changes when it must be generated internally versus when appropriate external support is supplied.

A cue rescue does not prove an 'enactment coupling' mechanism. Cue delivery can change alerting, expectancy, preparation, attention, or confidence. The final manuscript states that no independent noncircular proximal assay of F3 is currently established in this task family. Until one exists, F3 remains a reproducible experimental contrast whose stronger mechanistic interpretation is OPEN.

7. F4: is entering the target configuration itself costly?

Starting can require leaving one active configuration and entering another. Task-switching research already shows that switch trials are often slower or less accurate than repeat trials, and that preparation can reduce but not necessarily eliminate switching costs.

F4 therefore asks whether initiation is sensitive to the transition requirement. It is not a claim for a new localized control-state mechanism. A repeat-versus-switch effect can arise from reconfiguration, interference, inhibition, stimulus-response remapping, working-memory updating, preparation, or combinations of established processes.

If those established switching or control constructs fully explain the residual effect, the analysis must contract to them. F4 survives only as broad transition sensitivity to the extent the evidence still requires it. This preserves a useful distinction - difficulty entering a state need not equal difficulty continuing once there - without manufacturing a proprietary mechanism.

8. Post-initiation failure is not another kind of failure to start

A person can initiate successfully and then fail during execution. The manuscript marks that outcome separately as $Z^{\wedge}E$. Once the objective initiation criterion has been crossed, the trial cannot simultaneously count as $Y^{\wedge}I = 0$.

This sounds obvious, but it prevents a major attribution error. If a demanding task is started and then abandoned, stalls, or is performed poorly, that outcome may concern sustained effort, execution demand, skill, control production, fatigue, or another downstream process. It cannot be used retrospectively to redefine the initiation episode as though onset never occurred.

9. The ordered test sequence is designed to make the framework lose

Step	Question	Permitted conclusion	Forbidden shortcut
1. Establish G	Is the superordinate goal represented/endorsed?	The goal is present for the bounded episode.	Goal endorsement proves local policy selection.
2. Evaluate A / F1	Did action specification change?	A local next action became more determinate if the proximal A measure changes.	An initiation rescue proves an independent F1 cause.
3. Evaluate P / F2	Did the target policy win or gain choice propensity?	Local policy selection changed if the prespecified P measure supports it.	$P = 1$ guarantees onset.
4. Test F3	With A available and $P = 1$, does supported onset differ from self-generated onset?	A reproducible operational onset-support contrast exists.	Cue rescue identifies a unique mechanism.
5. Test F4	With onset support matched, do repeat and switch requirements differ?	Initiation is transition-sensitive at the operational level.	A switch cost proves a novel transition module.

Step	Question	Permitted conclusion	Forbidden shortcut
6. Measure $Z^{\wedge}E$	What happens after onset?	Execution outcome is characterized separately.	Post-initiation failure is reclassified as non-initiation.

The order is not a mandatory biological pipeline. Its purpose is to condition later contrasts on information needed to interpret them. The paper also warns that measuring or forcing P can itself alter commitment or preparation. An empirical design therefore has to minimize or model measurement reactivity rather than treating the act of checking the local choice as causally neutral.

10. F1 and F2 are allowed to collapse into one lower-dimensional account

One of the strongest scientific features of the final manuscript is that its factors are expendable. A supplied action plan can improve initiation because it genuinely changes action specification, because it raises efficacy or policy preference, or because those effects are inseparable under the available measures.

The framework therefore requires a prospective F1/F2 adjudication. If the effect of an action-specification manipulation is fully mediated by the measured change in P and an extra F1 term adds no held-out predictive value, the correct result is contraction: F1 does not survive as an independently identified factor in that task family.

That outcome is not 'weak evidence for F1.' It is evidence that the richer decomposition was unnecessary. The same rule applies elsewhere: added dimensions are empirical burdens, not entities the theory is obligated to preserve.

11. What the framework says about ADHD - and what it does not

Squirrel began from observations that are recognizable in ADHD discussions, but the final paper is not an ADHD mechanism paper. It includes ADHD and neurological evidence only where those literatures help demonstrate that valuation, intention, self-generated activation, and execution can come apart.

The manuscript does not claim that ADHD is caused by one initiation bottleneck, that people with ADHD share a uniformly high recruitment threshold, or that F1-F4 are ADHD traits. Individual episodes and people may show different patterns, and established lower-dimensional models can explain particular effects. The framework's contribution is to stop the endpoint 'did not start' from being used as if it already identified which process failed.

12. Model contraction, falsification, and invalid tests

The four labels form a hypothesis class, not a validated factorization. Future data can support separability, mediation, nesting, merger, dropping a contrast, a common-factor model, unstable person-level profiles, or invalidation of a proposed selective test.

- If a proposed F1 manipulation does not change the proximal action-specification measure A, it is not a valid F1 test.
- If F1 effects are fully carried by local policy selection and the richer model adds no held-out prediction, F1 and F2 should merge or contract.
- If self-generated versus supported onset does not survive valid control and replication, F3 should be dropped or merged.
- If ordinary switching, preparation, conflict, or remapping models fully account for repeat-versus-switch differences, use those models and do not preserve an independent F4 mechanism.
- If the candidate factors fit training manipulations but fail to predict held-out perturbations or cross-task structure, identification is not supported.
- If one lower-dimensional construct predicts the relevant initiation patterns as well as or better than the four-contrast architecture, the lower-dimensional model wins.

13. Conclusion

The paper does not answer 'why didn't you do it?' with another single label. It changes the question. If a goal was genuinely endorsed and the task still did not begin, which part of the goal-to-action transition remains ambiguous, and what intervention would distinguish the candidate explanations?

Action construction, local policy selection, self-generated versus supported onset, and transition-sensitive initiation are kept separate only while they earn that separation. The tests are ordered to prevent one rescue manipulation from being treated as a mechanism diagnosis. Proximal measures are required where possible, later contrasts are conditioned on earlier evidence, and post-initiation execution is kept outside the initiation endpoint.

The framework therefore offers an attribution discipline rather than a theory of character, motivation, ADHD, or willpower. Its success would be fewer invalid conclusions from the same visible endpoint. Its failure would be equally informative: if mature motivation and control models or a smaller factorization explain the data without the added distinctions, Squirrel should contract into them.

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